

The Period-4 Case of the Morton–Vivaldi Irreducibility Conjecture

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Abstract

Let Φ_k be the k -th cyclotomic polynomial, let φ denote Euler’s totient function, and define

$$g(T) = -T^4 - 2T^3 - 4T^2 - 6T + 5 + \frac{8}{T} + \frac{16}{T^2}.$$

For every $k \geq 2$, we prove that

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \Phi_k(g(T))$$

is irreducible over $\mathbf{Q}$. The proof combines reduction at primes above 2 with an archimedean bound for the roots. The normalization of the period-4 multiplier curve then yields the irreducibility of every period-4 delta factor $\Delta_{4k,4}$ for the quadratic family $z \mapsto z^2 + c$.

1 Introduction

For the quadratic family $f_c(z) = z^2 + c$, Morton and Vivaldi introduced delta factors whose roots are the parabolic parameters of prescribed formal and exact periods, and conjectured that all of these factors are irreducible over $\mathbf{Q}$ [MV95]. The cases of periods 1 and 2 reduce to cyclotomic irreducibility. The complete period-3 case was proved by Koizumi, Murakami, Sano, and Takehira [KMST25]. The first remaining complete period is therefore period 4.

Put $C = 4c$. The normalization of the period-4 multiplier curve is parametrized by

$$T \longmapsto (g(T), h(T)), \quad h(T) = -T^2 - 3 - \frac{4}{T}, \tag{1}$$

on $\mathbb{G}_m$; see [KMST25, Remark 2.10], where the parametrization is attributed to Giarrusso and Fisher [GF95]. The degree-six map g suggests pulling a cyclotomic polynomial back to the normalization. Although the displayed definition of Γ_k is initially Laurent, it is a monic polynomial in $\mathbf{Z}[T]$ of degree $6\varphi(k)$.

Our main result is the following.

Theorem 1.1. *For every integer $k \geq 2$, the polynomial $\Gamma_k(T)$ is irreducible over $\mathbf{Q}$.*

Together with the normalization of the multiplier curve, this settles the period-4 family of delta factors.

Corollary 1.2. *For every integer $k \geq 1$, the period-4 delta factor $\Delta_{4k,4}$ is irreducible over $\mathbf{Q}$.*

The argument has two complementary parts. At every prime above 2, a factorization modulo the prime ideal severely restricts the possible degrees of a global factor. Uniform complex root bounds then turn the remaining possibilities into contradictions involving algebraic norms.

2 The fixed-degree reformulation

Let ζ be a root of unity different from 1, and define

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16. \quad (2)$$

A direct calculation gives

$$g(T) - \zeta = -\frac{F_\zeta(T)}{T^2}. \quad (3)$$

Proposition 2.1. *Let $k \geq 2$, let ζ be a primitive k -th root of unity, and put $K = \mathbf{Q}(\zeta)$. The following conditions are equivalent:*

- (A) Γ_k is irreducible over $\mathbf{Q}$;
- (B) F_ζ is irreducible over K for one primitive k -th root ζ ;
- (C) F_ζ is irreducible over K for every primitive k -th root ζ .

Proof. The cyclotomic factorization and (3) give

$$\begin{aligned} \Gamma_k(T) &= (-1)^{\varphi(k)} T^{2\varphi(k)} \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} (g(T) - \zeta) \\ &= \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \end{aligned}$$

Thus Γ_k is monic of degree $6\varphi(k)$. Its coefficients are algebraic integers fixed by the Galois group of $\mathbf{Q}(\zeta)/\mathbf{Q}$, so they belong to $\mathbf{Z}$.

Let α be a root of F_ζ . Since $F_\zeta(0) = -16$, we have $\alpha \neq 0$, and (3) yields

$$\zeta = g(\alpha) \in \mathbf{Q}(\alpha).$$

Consequently $K \subseteq \mathbf{Q}(\alpha)$. If F_ζ is irreducible over K , then

$$[\mathbf{Q}(\alpha) : \mathbf{Q}] = [K(\alpha) : K][K : \mathbf{Q}] = 6\varphi(k).$$

The minimal polynomial of α over $\mathbf{Q}$ therefore has the same degree as Γ_k and divides it, so it equals Γ_k .

Conversely, suppose that Γ_k is irreducible. Then every root α of F_ζ has degree $6\varphi(k)$ over $\mathbf{Q}$. Since $K \subseteq \mathbf{Q}(\alpha)$, it follows that $[K(\alpha) : K] = 6$. The minimal polynomial of α over K has degree 6 and divides F_ζ , so F_ζ is irreducible. Finally, the automorphisms of $K/\mathbf{Q}$ act transitively on the primitive k -th roots and send F_ζ to the corresponding conjugate polynomial. This proves all three equivalences. $\square$

At $\zeta = 1$ one has the exceptional factorization

$$F_1(T) = (T + 2)(T^2 + 4)(T^3 - 2). \quad (4)$$

Set

$$P(T) = F_1(T).$$

Then

$$F_\zeta(T) = P(T) + (\zeta - 1)T^2. \quad (5)$$

3 A uniform archimedean bound

Lemma 3.1. *Let $|\zeta| = 1$. Every complex root α of F_ζ satisfies*

$$1 < |\alpha| < \frac{5}{2}.$$

Proof. Put $r = |\alpha|$. By (5),

$$|P(\alpha)| = |\zeta - 1|r^2 \leq 2r^2. \quad (6)$$

If $r \leq 1$, then the reverse triangle inequality and (4) give

$$|P(\alpha)| \geq (2 - r)(4 - r^2)(2 - r^3) \geq 3,$$

whereas $2r^2 \leq 2$, contradicting (6).

Now suppose that $r \geq 5/2$. Again by the reverse triangle inequality,

$$\frac{|P(\alpha)|}{r^2} \geq (r - 2)(r^2 - 4) \left(r - \frac{2}{r^2} \right).$$

Each of the three factors on the right is positive and increasing for $r \geq 5/2$. Hence

$$\frac{|P(\alpha)|}{r^2} \geq \frac{1}{2} \cdot \frac{9}{4} \cdot \frac{109}{50} = \frac{981}{400} > 2,$$

which also contradicts (6). The claimed strict bounds follow. $\square$

We shall use both sides of the annular bound. The constant term of any positive-degree monic factor of F_ζ has absolute value greater than 1 under every complex embedding, while each individual root has absolute value less than $5/2$.

4 Two factorization lemmas

Let K be a number field, let $\mathcal{O}_K$ be its ring of integers, and let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. Write $v_{\mathfrak{p}}$ for the additive valuation normalized by $v_{\mathfrak{p}}(K^\times) = \mathbf{Z}$. A bar denotes reduction modulo $\mathfrak{p}$.

Lemma 4.1. *If a monic polynomial $f \in \mathcal{O}_K[T]$ factors as $f = HJ$ with H, J monic in $K[T]$, then $H, J \in \mathcal{O}_K[T]$.*

Proof. Every root of f is integral over $\mathcal{O}_K$. The coefficients of H and J are elementary symmetric functions in subsets of these roots, hence are integral over $\mathcal{O}_K$. They also lie in K . Since $\mathcal{O}_K$ is integrally closed in K , they belong to $\mathcal{O}_K$. $\square$

Lemma 4.2 (the unsplit-block lemma). *Let $f \in \mathcal{O}_K[T]$ be monic and suppose that, for some integers $a, b \geq 0$ and some $t \in \mathcal{O}_K$ with $\bar{t} \neq 0$,*

$$\bar{f}(T) = T^a(T - \bar{t})^b, \quad f(t) \in \mathfrak{p} \setminus \mathfrak{p}^2.$$

If $f = HJ$ with $H, J \in \mathcal{O}_K[T]$ monic and $\deg H < b$, then

$$\bar{H}(T) = T^{\deg H} \quad \text{and} \quad \deg H \leq a.$$

Proof. Unique factorization over the residue field gives integers r, s such that

$$\bar{H} = T^r(T - \bar{t})^s, \quad \bar{J} = T^{a-r}(T - \bar{t})^{b-s}.$$

Because H and J are monic, reduction preserves their degrees. If $s > 0$, then $s \leq \deg H < b$, so both s and $b - s$ are positive. Consequently $H(t), J(t) \in \mathfrak{p}$, which would imply $f(t) = H(t)J(t) \in \mathfrak{p}^2$. This is impossible. Thus $s = 0$, so $\bar{H} = T^r$. Degree preservation gives $r = \deg H \leq a$. $\square$

We also use the standard elementary fact that if a sum in a discretely valued field is zero, then its terms cannot have a unique least valuation.

5 Cyclotomic facts at the prime 2

Let ζ have order

$$k = 2^a m, \quad m \text{ odd},$$

and put $K = \mathbf{Q}(\zeta)$. There is a unique decomposition

$$\zeta = \xi\eta, \tag{7}$$

where ξ has order 2^a and η has order m .

We recall the local cyclotomic facts used below. If $\mathfrak{p} \mid 2$, then reduction is injective on roots of unity of odd order, while every root of unity of 2-power order reduces to 1. The odd-order cyclotomic extension is unramified at 2. If $a \geq 2$, then $\xi - 1$ is a uniformizer and

$$v_{\mathfrak{p}}(2) = 2^{a-1}. \tag{8}$$

Indeed, with $n = 2^{a-1}$, the element $\xi - 1$ is a root of

$$(1 + X)^n + 1,$$

which is Eisenstein at 2. Passing to the compositum with the unramified odd-order part preserves the uniformizer and the ramification index. When $a = 0$, the extension is unramified at 2; when $a = 1$ and m is odd, $\mathbf{Q}(\zeta_{2m}) = \mathbf{Q}(\zeta_m)$, so it is again unramified at 2. These statements are also immediate from the usual decomposition law for primes in cyclotomic fields; see, for example, [Was97, Chapter 2].

6 Orders with a nontrivial odd part

Assume throughout this section that $m > 1$. Fix a prime $\mathfrak{p} \mid 2$ and write

$$e = v_{\mathfrak{p}}(2).$$

Since 4 is invertible modulo m , there is an odd-order root of unity $\theta \in K$ such that $\theta^4 = \eta$. Put

$$t_0 = \theta - 1.$$

The reduction of θ is not 1, so t_0 is a $\mathfrak{p}$ -adic unit.

Since $\bar{\xi} = 1$ and $\bar{\eta} \neq 1$, reduction of (2) in characteristic 2 gives

$$\begin{aligned} \overline{F_\zeta(T)} &= T^6 + (\bar{\eta} + 1)T^2 \\ &= T^2((T+1)^4 + \bar{\eta}) \\ &= T^2(T - \bar{t}_0)^4. \end{aligned} \tag{9}$$

Lemma 6.1. *For every prime $\mathfrak{p} \mid 2$,*

$$v_{\mathfrak{p}}(F_\zeta(t_0)) = 1.$$

Proof. Define the Laurent polynomial

$$B(T) = T^4 + 3T^3 + 5T^2 + 5T - 2 - \frac{4}{T} - \frac{8}{T^2}.$$

The identity

$$g(T) - (T+1)^4 = -2B(T)$$

and $(t_0 + 1)^4 = \eta$ imply

$$F_\eta(t_0) = 2t_0^2 B(t_0). \tag{10}$$

Modulo $\mathfrak{p}$,

$$\bar{B}(T) = T^4 + T^3 + T^2 + T = T(T+1)^3.$$

This reduction is legitimate because t_0 is a $\mathfrak{p}$ -adic unit. Therefore

$$\bar{B}(t_0) = \bar{t}_0 \bar{\theta}^3 \neq 0,$$

and (10) gives $v_{\mathfrak{p}}(F_\eta(t_0)) = e$.

Moreover,

$$F_\zeta(T) = F_\eta(T) + \eta(\xi - 1)T^2. \tag{11}$$

If $a = 0$, then $\xi = 1$ and $e = 1$, so the assertion follows from (10). If $a \geq 2$, the two terms on the right side of (11), evaluated at t_0 , have valuations $e \geq 2$ and 1, respectively, so their sum has valuation 1.

It remains to consider $a = 1$. Here $\xi = -1$ and $e = 1$, whence

$$F_\zeta(t_0) = 2t_0^2(B(t_0) - \eta).$$

After division by 2 and reduction modulo $\mathfrak{p}$,

$$\begin{aligned} \frac{\overline{F_\zeta(t_0)}}{2} &= \bar{t}_0^2(\bar{t}_0 \bar{\theta}^3 + \bar{\theta}^4) \\ &= \bar{t}_0^2 \bar{\theta}^3 \neq 0. \end{aligned}$$

Thus the valuation is 1 in every case. $\square$

Proposition 6.2. *If the order of ζ has a nontrivial odd part, then F_ζ is irreducible over K .*

Proof. Suppose that F_ζ is reducible over K . Choose a monic irreducible factor H of minimal degree d , and write $F_\zeta = HJ$ with J monic. Since $\deg F_\zeta = 6$, we have $d \leq 3$. By Lemma 4.1, both factors lie in $\mathcal{O}_K[T]$.

At each prime $\mathfrak{p} \mid 2$, equations (9) and Lemma 6.1 allow us to apply Lemma 4.2 with $(a, b) = (2, 4)$. Since $d < 4$, it follows that $d \leq 2$ and

$$\bar{H} = T^d. \quad (12)$$

First suppose that $d = 2$. Put

$$b = H(0), \quad c = J(0).$$

Then $bc = -16$. At every $\mathfrak{p} \mid 2$, (12) and (9) give

$$\bar{J} = (T - \bar{t}_0)^4,$$

so c is a $\mathfrak{p}$ -adic unit. At primes not above 2, the integrality of b, c and $bc = -16$ also show that c is a unit. Hence c is a unit of $\mathcal{O}_K$.

For every complex embedding $\sigma : K \hookrightarrow \mathbf{C}$, the polynomial $\sigma(J)$ is a monic factor of $F_{\sigma(\zeta)}$ of degree 4. By Lemma 3.1, each of its roots has modulus greater than 1. Thus $|\sigma(c)| > 1$ for every σ , and therefore

$$|\mathrm{N}_{K/\mathbf{Q}}(c)| = \prod_{\sigma} |\sigma(c)| > 1.$$

This contradicts the fact that the norm of a unit is ± 1 .

Now suppose that $d = 1$, say $H(T) = T - \alpha$. From (12), $q = v_{\mathfrak{p}}(\alpha) > 0$ for every $\mathfrak{p} \mid 2$. Moreover, $\zeta - 5$ is a $\mathfrak{p}$ -adic unit, because $\bar{\zeta} = \bar{\eta} \neq 1$. The valuations of the seven terms in $F_\zeta(\alpha) = 0$, ordered from the constant term to the leading term, are

$$4e, \quad 3e + q, \quad 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q < 2e$, then $2q$ is the unique least value. If $q > 2e$, then $4e$ is the unique least value. Since a zero sum cannot have a unique term of least valuation, necessarily

$$v_{\mathfrak{p}}(\alpha) = 2e = v_{\mathfrak{p}}(4) \quad (\mathfrak{p} \mid 2). \quad (13)$$

At every prime away from 2, the equality $(-\alpha)J(0) = -16$ and integrality give valuation zero for α . Thus $(\alpha) = (4)$, and $u = \alpha/4$ is a unit of $\mathcal{O}_K$.

For every complex embedding σ , Lemma 3.1 gives

$$|\sigma(u)| = \frac{|\sigma(\alpha)|}{4} < \frac{5}{8}.$$

It follows that $|\mathrm{N}_{K/\mathbf{Q}}(u)| < 1$, contradicting that u is a unit. Both possible degrees have been excluded, so F_ζ is irreducible. $\square$

7 Pure powers of 2

Assume now that ζ has order 2^a with $a \geq 1$. There is a unique prime $\mathfrak{p}$ of K above 2, its residue degree is 1, and

$$e = v_{\mathfrak{p}}(2) = \varphi(2^a).$$

For $a = 1$, $K = \mathbf{Q}$ and $\zeta = -1$. For $a \geq 2$, $\zeta - 1$ is a uniformizer. In all cases,

$$v_{\mathfrak{p}}(\zeta - 5) = 1. \quad (14)$$

Indeed, for $a = 1$ one has $\zeta - 5 = -6$; for $a \geq 2$ one has $\zeta - 5 = (\zeta - 1) - 4$, whose two summands have valuations 1 and $2e > 1$.

Reduction modulo $\mathfrak{p}$ gives

$$\overline{F_{\zeta}(T)} = T^6. \quad (15)$$

Proposition 7.1. *If ζ has 2-power order and $\zeta \neq 1$, then F_{ζ} is irreducible over K .*

Proof. Suppose that F_{ζ} is reducible. Choose a monic irreducible factor H of minimal degree $d \leq 3$, and write $F_{\zeta} = HJ$ with J monic. Lemma 4.1 gives $H, J \in \mathcal{O}_K[T]$, and (15) gives

$$\bar{H} = T^d, \quad \bar{J} = T^{6-d}. \quad (16)$$

We first exclude $d = 1$. Write $H = T - \alpha$. Then $q = v_{\mathfrak{p}}(\alpha)$ is a positive integer. The valuations of the terms of $F_{\zeta}(\alpha) = 0$, from constant to leading, are

$$4e, \quad 3e + q, \quad 1 + 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q \leq 2e - 1$, then $1 + 2q$ is the unique least value. If $q \geq 2e$, then $4e$ is the unique least value. Both cases contradict the nonarchimedean triangle inequality. Hence $d \neq 1$.

Next suppose that $d = 3$. Write

$$H = T^3 + h_2T^2 + h_1T + h_0, \quad J = T^3 + j_2T^2 + j_1T + j_0.$$

By (16), every displayed coefficient other than the leading coefficients belongs to $\mathfrak{p}$. The coefficient of T^2 in HJ is

$$h_2j_0 + h_1j_1 + h_0j_2 \in \mathfrak{p}^2.$$

But the coefficient of T^2 in F_{ζ} is $\zeta - 5$, which belongs to $\mathfrak{p} \setminus \mathfrak{p}^2$ by (14). This contradiction excludes $d = 3$.

Consequently $d = 2$. Write

$$H = T^2 + h_1T + h_0, \quad J = T^4 + j_3T^3 + j_2T^2 + j_1T + j_0.$$

All the h_i and j_i lie in $\mathfrak{p}$. Comparing the coefficient of T^2 gives

$$\zeta - 5 = j_0 + h_1j_1 + h_0j_2 \equiv j_0 \pmod{\mathfrak{p}^2}. \quad (17)$$

Therefore $j_0 \in \mathfrak{p} \setminus \mathfrak{p}^2$.

Put $b = H(0)$ and $c = J(0) = j_0$. At every odd prime, the integrality of b, c and $bc = -16$ implies that c is a unit. Since $\mathfrak{p}$ is the unique prime above 2, (17) shows that

$$(c) = \mathfrak{p}.$$

The residue degree of $\mathfrak{p}$ is 1, so

$$|N_{K/\mathbf{Q}}(c)| = N(\mathfrak{p}) = 2. \quad (18)$$

For every complex embedding σ , the two roots of $\sigma(H)$ have modulus less than $5/2$ by Lemma 3.1. Since the absolute value of the product of all six roots of $F_{\sigma(\zeta)}$ is 16,

$$|\sigma(c)| = \frac{16}{|\sigma(b)|} > \frac{16}{(5/2)^2} = \frac{64}{25}.$$

Taking the product over all embeddings gives

$$|N_{K/\mathbf{Q}}(c)| > \left(\frac{64}{25}\right)^{[K:\mathbf{Q}]} > 2,$$

contradicting (18). Thus F_ζ is irreducible. $\square$

Propositions 6.2 and 7.1 prove that F_ζ is irreducible over $\mathbf{Q}(\zeta)$ for every root of unity $\zeta \neq 1$. Proposition 2.1 now proves Theorem 1.1.

8 Passage to the period-4 delta factors

We use the standard notation of [Mor96, MV95, KMST25]. Let $\delta_m(x, c)$ denote the multiplier polynomial for cycles of formal period m , using the normalization that becomes monic in $C = 4c$, and put

$$\tilde{\delta}_m(x, C) = \delta_m(x, C/4).$$

The multiplier curve X_m is the affine curve

$$\tilde{\delta}_m(x, C) = 0.$$

For $k \geq 2$, define

$$\tilde{\Delta}_{mk,m}(C) = \Delta_{mk,m}(C/4) = \text{Res}_x(\Phi_k(x), \tilde{\delta}_m(x, C)).$$

This is [KMST25, Equation (1.4)] after the change of variable $C = 4c$.

We recall three facts from [KMST25]. First, $\tilde{\Delta}_{mk,m}$ is separable [KMST25, Proposition 2.3]. Second, writing $Z(q)$ for the set of roots of a polynomial q in $\overline{\mathbf{Q}}$ without multiplicity, projection onto the parameter coordinate gives the following Galois-equivariant bijection [KMST25, Equation (2.2)]:

$$\{(\lambda, C) \in X_m(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\} \xrightarrow{\sim} Z(\tilde{\Delta}_{mk,m}). \quad (19)$$

Third, [KMST25, Equation (1.6)] gives

$$\deg_C \tilde{\Delta}_{mk,m} = \nu(m)\varphi(k), \quad \nu(m) = \sum_{d|m} 2^{d-1} \mu(m/d). \quad (20)$$

Here μ denotes the Möbius function. In particular, $\nu(4) = 0 - 2 + 8 = 6$.

For $m = 4$, the normalization is precisely the finite surjective map (1); see [KMST25, Remark 2.10]. Fix $k \geq 2$. Since $\Gamma_k(0) = (-16)^{\varphi(k)}$, all of its roots lie in $\mathbb{G}_m$. Restricting the normalization map to the roots of Γ_k gives a surjection

$$Z(\Gamma_k) \longrightarrow \{(\lambda, C) \in X_4(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\}. \quad (21)$$

Indeed, normalization is finite and surjective, and every preimage of a point on the right has nonzero parameter T and satisfies $\Phi_k(g(T)) = 0$. Conversely, every root of Γ_k yields such a point. By separability, (19), and (20), the target has exactly $6\varphi(k)$ elements. The source has at most $6\varphi(k)$ elements because this is the degree of Γ_k . The displayed surjection therefore forces both sets to have $6\varphi(k)$ elements and is a Galois-equivariant bijection.

Combining it with (19) gives a Galois-equivariant bijection

$$Z(\Gamma_k) \xrightarrow{\sim} Z(\tilde{\Delta}_{4k,4}).$$

The absolute Galois group acts transitively on the left by Theorem 1.1, hence transitively on the right. Since $\tilde{\Delta}_{4k,4}$ is separable, it is irreducible over $\mathbf{Q}$.

It remains to treat $k = 1$. After the substitution $C = 4c$, the table in [MV95, Section 3, Table 1] gives

$$\tilde{\delta}_4(1, C) = (C^2 - 2C + 5)(C + 5)(C^3 + 9C^2 + 27C + 135).$$

For completeness, $\tilde{\delta}_1(x, C) = x^2 - 2x + C$ and $\tilde{\delta}_2(x, C) = C + 4 - x$; the resultant definitions therefore give

$$\tilde{\Delta}_{4,1} = C^2 - 2C + 5, \quad \tilde{\Delta}_{4,2} = C + 5.$$

Thus, in the $C = 4c$ normalization, the exact period-4 factor is

$$\tilde{\Delta}_{4,4}(C) = C^3 + 9C^2 + 27C + 135 = (C + 3)^3 + 108.$$

A rational root would be an integer and would make an integer cube equal to -108 . This is impossible because $v_2(108) = 2$ is not divisible by 3. Thus this cubic is irreducible. Finally, the invertible linear change $C = 4c$ preserves irreducibility over $\mathbf{Q}$. This proves Corollary 1.2.

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